



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 01

**Experiment Title: Design of CMOS NOR Gate Circuit in Cadence Virtuoso
and Analysis of Transient and DC Response**

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1. Experiment Name

Design of CMOS NOR Gate Circuit in Cadence Virtuoso and Analysis of Transient and DC Response

2. Objective

- To design a 2-input CMOS NOR gate using Cadence Virtuoso.
- To create the schematic and simulation testbench of the NOR gate.
- To perform transient analysis and observe the input-output behavior of the NOR gate.
- To perform DC analysis and study the voltage transfer characteristics of the NOR gate.
- To verify that the simulated output follows the expected NOR gate logic operation.

3. Apparatus / Hardware & Software Requirements

- Cadence Virtuoso Studio – for schematic design and circuit simulation.
- ADE Explorer – for transient and DC analysis.
- Simulator Spectre – used as the circuit simulator for running the analyses.
- CMOS transistor devices – two PMOS and two NMOS transistors for implementing the 2-input NOR gate.
- Voltage sources – used for the supply voltage and input signals in the simulation testbench.
- Computer/Workstation with access to the Cadence environment.

4. Circuit Diagram / Schematic

The 2-input CMOS NOR gate was designed using two PMOS transistors connected in series in the pull-up network and two NMOS transistors connected in parallel in the pull-down network. The circuit has two inputs, V_a and V_b , and one output, V_{out} .

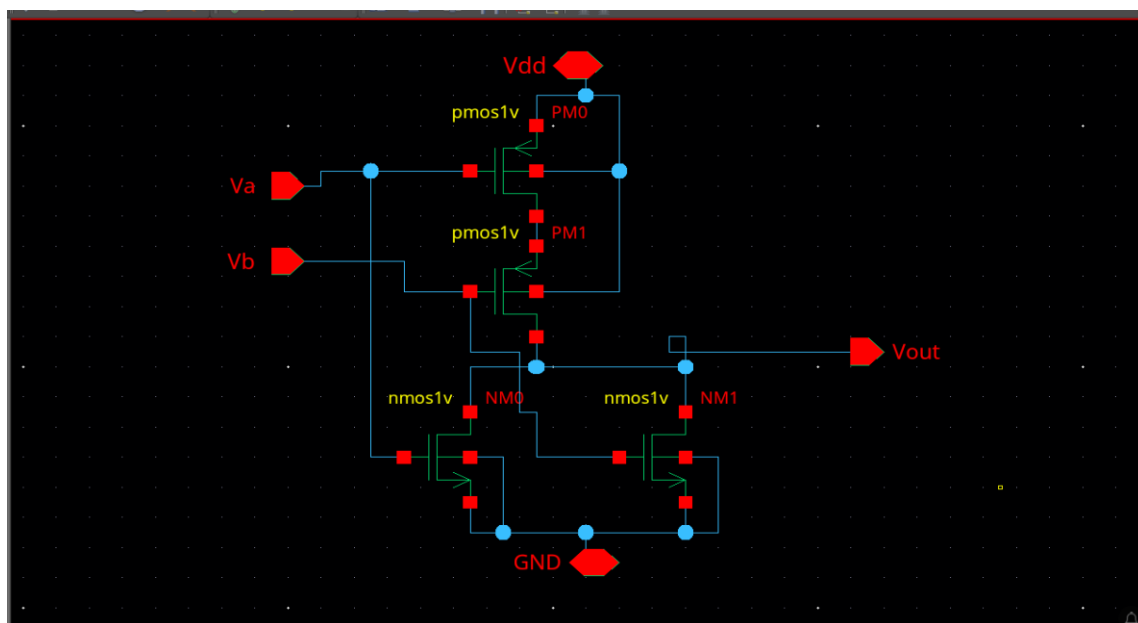


Figure 1: Transistor-Level Schematic of the 2-Input CMOS NOR Gate

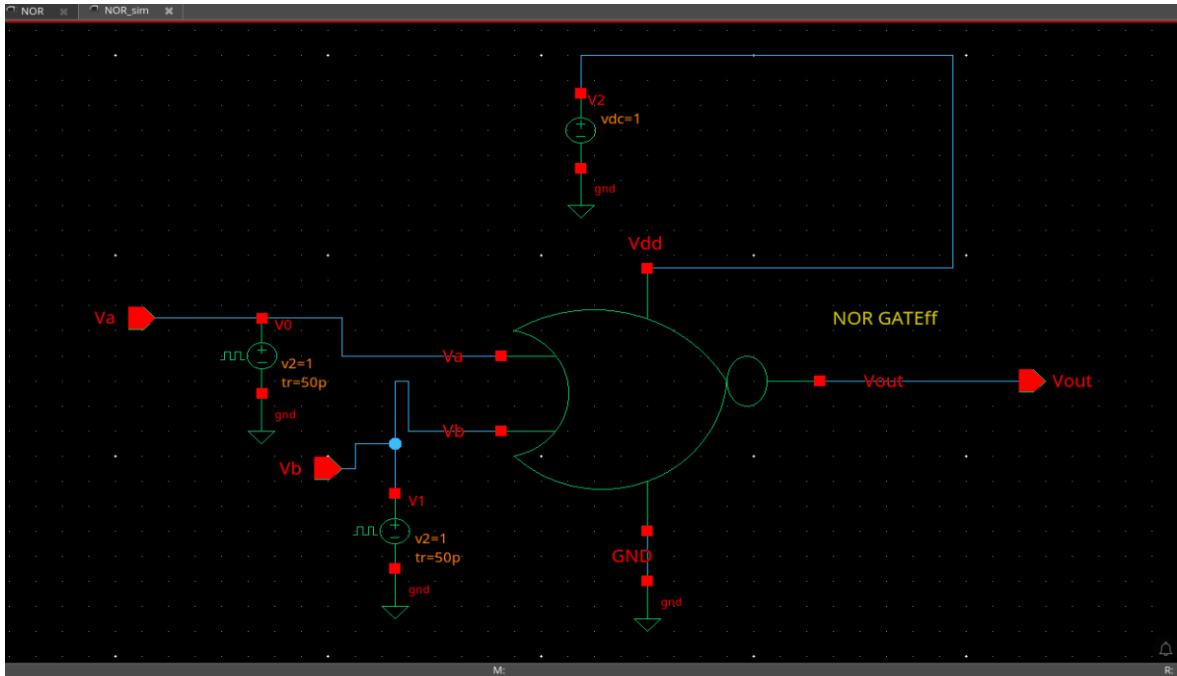


Figure 2: Simulation Testbench of the 2-Input CMOS NOR Gate

6. Output Analysis / Observations

Transient Analysis:

The transient simulation shows the input signals V_a and V_b and the corresponding output V_{out} . The output becomes HIGH only when both inputs are LOW. For all other input combinations, the output remains LOW. This behavior matches the expected truth table of a 2-input NOR gate.

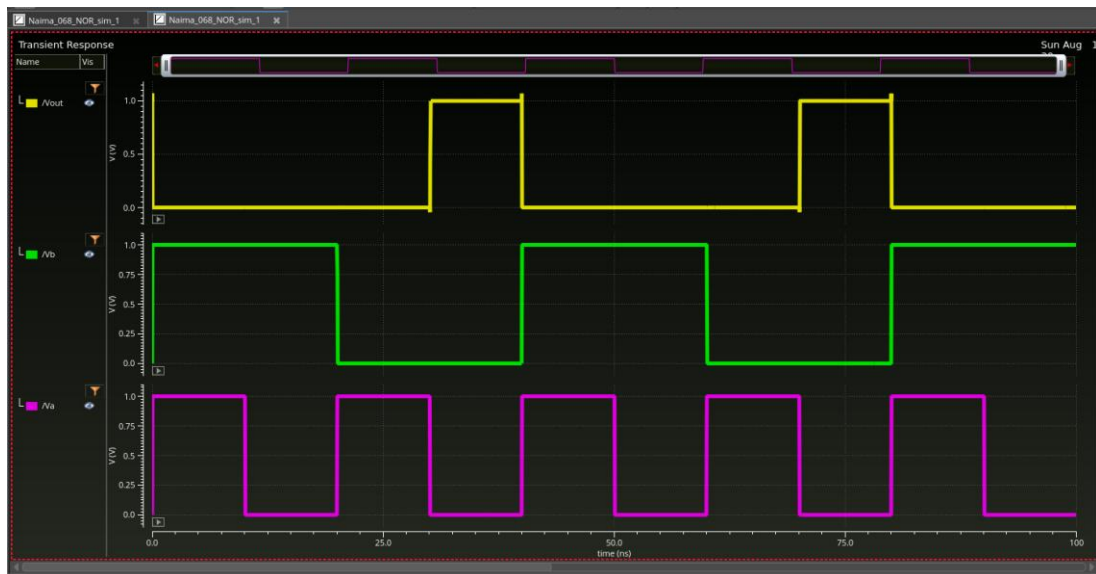


Figure 3: Transient Response of the 2-Input CMOS NOR Gate

DC Analysis:

The DC response shows the variation of the output voltage with respect to the swept input voltage. The output remains close to V_{dd} when the input voltage is low and decreases toward 0 V as the input voltage increases. A sharp transition is observed around the switching region, which indicates the expected voltage transfer characteristic of the CMOS NOR gate when the other input is held LOW.



Figure 4: DC Response of the 2-Input CMOS NOR Gate

The truth table of the 2-input CMOS NOR gate is shown below.

Va	Vb	Vout	Observation
0	0	1	HIGH
0	1	0	LOW
1	0	0	LOW
1	1	0	LOW

Table 1: Truth Table of the 2-Input CMOS NOR Gate

7. Result

The 2-input CMOS NOR gate was successfully designed and simulated using Cadence Virtuoso. The transient analysis showed that the output becomes HIGH only when both inputs are LOW, and remains LOW for all other input combinations. The DC analysis also showed the expected voltage transfer behavior of the NOR gate. Therefore, the simulated results matched the theoretical operation of a 2-input NOR gate, and the objectives of the experiment were successfully achieved.

8. Github Repository Commits

- **Commit:** Add NOR gate schematics and simulation results
- **Commit Link:** <https://github.com/naima-tabassum/CSE450-VLSI-Lab/commit/c4a52757c122cb4f9ac40b738de883300d1e7872>

9. Conclusion

This experiment provided practical experience in designing and simulating a 2-input CMOS NOR gate using Cadence Virtuoso. The operation of the pull-up and pull-down transistor networks was understood through the schematic design. Transient analysis was used to verify the NOR logic behavior, while DC analysis was used to observe the voltage transfer characteristic of the circuit. Overall, the experiment improved the understanding of CMOS NOR gate design and simulation using Cadence Virtuoso.